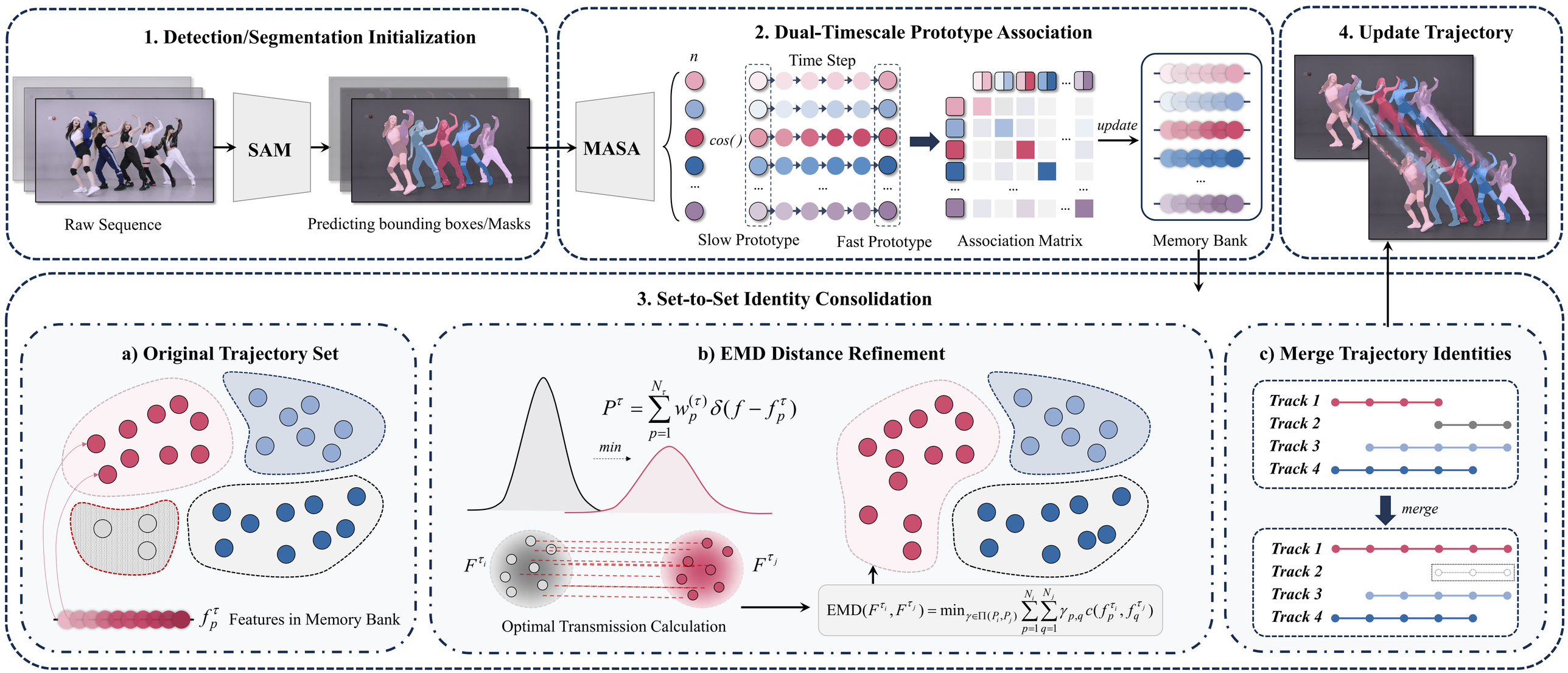
1. Detection/Segmentation Initialization
Raw Sequence
SAM
Predicting bounding boxes/Masks
MASA
2. Dual-Timescale Prototype Association
n
cos( )
Time Step
Slow Prototype
Fast Prototype
Association Matrix
update
Memory Bank
4. Update Trajectory
3. Set-to-Set Identity Consolidation
a) Original Trajectory Set
f_p^τ Features in Memory Bank
b) EMD Distance Refinement
$P^\tau = \sum_{p=1}^{N_\tau} w_p^{(\tau)} \delta(f - f_p^\tau)$
min
F^{τ_i}
F^{τ_j}
Optimal Transmission Calculation
$\mathrm{EMD}(F^{\tau_i}, F^{\tau_j}) = \min_{\gamma \in \Pi(P_i, P_j)} \sum_{p=1}^{N_i} \sum_{q=1}^{N_j} \gamma_{p,q} c(f_p^{\tau_i}, f_q^{\tau_j})$
c) Merge Trajectory Identities
Track 1
Track 2
Track 3
Track 4
merge
Track 1
Track 2
Track 3
Track 4